DBS Project
Instructions for Students

1. Project Objective

Design and develop a small database-driven application based on a real-world problem. The project should demonstrate how a database is designed, implemented, and integrated with an application to provide meaningful functionality.

2. Choose a Real-World Problem / Domain

Select a manageable application domain, for example:

- Library Management System
- Hospital / Clinic Management System
- Food Ordering System
- Student Course Registration System
- Hostel Management System
- Event Management System
- Vehicle / Parking Management System
- Inventory Management System
- Online Appointment System

Choose a problem that is sufficiently small to complete within the project period, but large enough to demonstrate database design and application development.

3. Identify Functional Modules

Break the application into meaningful functional modules. A module represents a functionality of the application, not simply a database table.

For example, a Library Management System may contain:

- Student/User Management
- Book Management
- Book Issue / Return
- Fine Management
- Search / Reports

One module may use multiple tables. For example, the Book Issue/Return module may involve Student, Book, Book_Copy, Issue and Fine tables.

4. Define the Problem and Objectives

- Clearly state the problem being addressed.
- Explain why the application is required.
- List the main objectives of the system.
- Clearly identify the users of the application, where applicable.

5. Identify Entities, Attributes and Relationships

- Identify the major entities involved in the application.
- Identify the attributes of each entity.
- Identify relationships among the entities.
- Identify primary keys and foreign keys.
- Identify appropriate constraints such as NOT NULL, UNIQUE, CHECK and referential integrity constraints.

6. Design the ER Diagram

Prepare an Entity-Relationship (ER) diagram representing the entities, attributes, relationships and relevant cardinalities of the system.

7. Convert the Design into a Relational Schema

- Convert the ER design into relational tables.
- Clearly specify primary keys and foreign keys.
- Show important constraints.
- Ensure that the schema represents the requirements of the application.

8. Implement the Database

- Create the database and tables using SQL.
- Insert realistic sample data.
- Implement required constraints.
- Perform necessary INSERT, UPDATE and DELETE operations.
- Verify that the database correctly supports the application requirements.

9. Develop the Application / User Interface

The project should include a working application/interface through which users can interact with the database. The interface does not need to be commercially sophisticated; it should be functional and demonstrate database interaction.

Possible technology combinations include:

- Python + PostgreSQL
- Java + PostgreSQL
- PHP + PostgreSQL
- Node.js + PostgreSQL
- Another suitable application technology approved for the project

10. Demonstrate Database Operations

The application should demonstrate meaningful operations such as:

- Adding new records
- Updating existing records
- Deleting records where appropriate
- Searching and retrieving records
- Displaying relevant information
- Generating useful reports or summaries

11. SQL Queries

Include meaningful SQL queries that demonstrate the database concepts covered in the course, as applicable to the selected project.

- SELECT and filtering using WHERE
- ORDER BY
- Aggregate functions such as COUNT, SUM, AVG, MIN and MAX
- GROUP BY and HAVING
- Joins
- Subqueries
- INSERT, UPDATE and DELETE

- Views, where appropriate

Use only the SQL concepts that are relevant to the selected application.

12. Normalization

Design the relations appropriately and apply normalization principles where applicable. The final design should avoid unnecessary redundancy and update anomalies.

13. Reports / Outputs

The application should provide useful outputs based on the project requirements. Examples include issue/return reports, appointment lists, order summaries, inventory status, student registration details, or other domain-specific reports.

14. Project Documentation

The project documentation should contain, as applicable:

- Project title
- Problem statement
- Objectives
- Description of the application
- Functional modules
- ER diagram
- Relational schema
- Important SQL commands and queries
- Application screenshots / outputs
- Description of how the application interacts with the database
- Results and observations

15. Project Demonstration

During the demonstration, students should be able to show:

- The working application.
- The major functional modules.
- Database tables and relationships.
- How the application performs database operations.
- Representative SQL queries and their outputs.
- Important reports or results generated by the system.

16. Important Points

- The project is a database-driven application, not just a set of SQL queries.
- The application should actually interact with the database.
- The database design should support the functional requirements of the application.
- Use realistic and meaningful sample data.
- Keep the design, database and application consistent with one another.

17. Recommended Project Workflow

1. Select a real-world problem/domain.
2. Identify users and requirements.
3. Identify functional modules.
4. Identify entities, attributes and relationships.
5. Prepare the ER diagram.

6. Convert the design into relational tables.
7. Implement the database using SQL.
8. Insert realistic sample data.
9. Develop the application/interface.
10. Connect the application to the database.
11. Implement required database operations and queries.
12. Generate useful outputs/reports.
13. Test the application.
14. Prepare documentation and demonstrate the project.

Key Requirement: You are expected to design and develop an application that uses a database to provide meaningful functionality. The project should demonstrate both database design and application development.